Thematic Analysis of Educational Activity Design

Researcher Note: Activities highlighted in green were the ones evaluated by subject matter experts.

1. RAG (Retrieval-Augmented Generation) Condition

Round	Activity Status	Interaction Summary	Significant Modification to the activity generated by AI?	Time (min)	Interactions	Format corrections or extensions once the activity was generated
1	Incomplete	Initiated session with a predefined physical interaction concept and requested a narrative. Selected an AI idea and made iterative requests to adapt formatting, context, and objectives specifically for the Protobject platform.	No (Formatting only)	DNF	7	4
2	Complete (Invalid)	No associated chat log available for analysis.	N/A	N/A	N/A	N/A
3	Complete	Requested curricular themes suitable for	No (Formatting	17	9	5

		specific devices. Evaluated two ideas in parallel before selecting one. Demanded template alignment, deeper content integration, and relevant study questions.	only)			
1	Complete	Began with a complete physical interaction concept and requested a narrative. Selected an idea and asked for the block-based solution in Protobject (text and image formats).	No (Formatting only)	13	5	2
2	Complete	Provided a clear pedagogical objective and requested a corresponding activity. Selected one idea and focused on simplifying the instructions for the students.	No (Formatting only)	5	4	1
3	Complete	Requested activities with minimal parameters, then explicitly redirected the AI to design a "noise traffic light". Provided a template image, modified the device functionality (intensity indicator without saving), and iteratively refined instructions and extension challenges.	Yes	18	10	6

1	Complete	Requested an activity for IB Computer Science. Rejected the AI's pseudocode approach, successfully requesting a version where students develop the core logic themselves.	No (Formatting only)	11	4	1
2	Complete	Started without parameters. Selected an idea to fit a specific template. Iteratively forced significant changes: added objectives, rewrote the narrative context entirely, demanded clear implementation instructions, and tailored the extension for Protobject.	Yes	17	8	5
3	Complete	Attached a guide document directly, bypassing initial questions. Re-attached the template after selecting an idea to force completion. Added learning objectives and exported final documents to Drive.	No (Formatting only)	13	7	1
1	Complete	Provided the evaluation unit and template items. Prompted for non-nature related ideas. Selected	No (Formatting only)	13	5	2

		one, modified the context slightly, and requested age-adapted, step-by-step instructions.				
2	Complete	Uploaded a template image requesting a general physics activity. Iterated through multiple batches of ideas. Selected an idea and engaged in a prolonged formatting struggle, repeatedly re-uploading the template and pointing out missing elements (age, objectives, devices) before finalizing extensions.	No (Formatting only)	22	13	8
3	Complete	Initiated the query by searching for a specific learning objective rather than a direct activity. Answered guiding questions and accepted a proposed activity without major changes.	No (Formatting only)	7	3	0
1	Almost Complete	Provided a specific math objective and requested a guide utilizing all Bloom's taxonomy skills. Ultimately discarded the AI's generated activity, choosing to manually fill the template with their own pre-existing idea.	N/A (Replaced by user)	10	4	0

2	Complete	Bypassed questions, providing full context for a "radar for visually impaired people". Constrained sensor usage. Requested Spanish pseudocode, engaged in unrelated philosophical/technical chat, attempted to erase chat history, and finally modified the activity target to deaf students.	Yes	18 (5 effective for the activity)	19 (5 effective for the activity)	2
3	Complete	Requested a highly practical, non-decorative activity. Corrected the AI's content focus and significantly refined the scientific methodology (e.g., using standard deviation instead of noise averages to measure sound quality). Iteratively humanized the narrative and tailored it for exam preparation.	Yes	17	17	14
1	Complete	Specified course, devices, and a kinematics focus. Corrected device terminology. Adjusted the narrative for a specific school context and requested detailed, enumerated implementation steps for	Yes	12	7	4

		students.				
2	Empty	Requested a vocational orientation activity. Did not progress past the initial idea generation phase or complete the template.	N/A	N/A	N/A	N/A
3	Complete	Answered questions and successfully developed a single idea without major alterations.	No (Formatting only)	7	3	0
1	Complete	Utilized an external non-gem chat to define an IB ESS objective, then transferred it to the gem for activity design. Requested clear Protobject instructions and re-aligned the AI when the original idea started drifting.	No (Formatting only)	13	8	5
2	Complete	Answered questions, rejected the first set of ideas, selected an idea from a second proposed batch, and requested exact template formatting.	No (Formatting only)	6	5	1
3	Empty	Bypassed the ideation phase, directly demanding a completed class template multiple	N/A	N/A	N/A	N/A

		times without success.				

2. Standard Condition

Round	Activity Status	Interaction Summary	Significant Modification to the activity generated by AI?	Time (min)	Interactions	Format corrections or extensions once the activity was generated
1	Complete	Selected an idea and engaged in extensive formatting iteration. Questioned the difficulty classification, constrained device usage, and repeatedly requested simplified, grade-appropriate language for explanations.	No (Formatting only)	4	12	9
2	Complete	Requested unconstrained brainstorming for math topics. Rejected all 5 initial proposals, changed the target content, and manually provided the core device usage concept. Forced the AI to develop this specific	Yes	16	7	4

		concept, demanding deep content focus over programming.				
3	Complete	Requested an activity, answered questions, and selected an idea without significant alterations.	No	10	3	0
1	Complete (Invalid)	Chat log missing. Activity appears AI-generated but may not have utilized the designated system.	N/A	N/A	N/A	N/A
2	Complete	Answered questions, selected an idea, requested a minor technical adjustment regarding light levels, and requested supplementary data on standard luminosity values.	No (Formatting only)	14	5	1
3	Empty	Answered questions and generated an idea, but failed to complete the activity template.	N/A	N/A	N/A	N/A
1	Generates 3 Complete Activities (Only first activity considered	Selected an idea and heavily directed its development (Spanish commands, sensor details). Iteratively formatted the output. Successfully requested variations for visually	Yes (Significant modifications on A1)	20 (10/5/5)	13 (11/1/1)	8 (A1 only)

	for analysis)	impaired students (Activity 2), then merged designs into a fully accessible prototype (Activity 3).				
2	Complete	Rejected math topics, pivoted to "urban music". Constrained to Protobject only. The teacher heavily modified the selected idea to restrict device usage, realigned the objective to explicitly target Computational Thinking pillars through problem-solving, and requested cultural adaptation (Chilean school festival narrative).	Yes	12	13	6
3	Complete	Utilized a secondary non-gem AI to cross-evaluate and filter ideas generated by the main system across intermediate and advanced levels. Copied the optimal idea back to the gem for narrative refinement and formatting. Evaluated the final result with the secondary AI.	No (Formatting only)	7	7	3
1	Complete	Uploaded template image requesting an activity aligned with the national physics	No (Formatting only)	7	4	1

		curriculum. Selected an idea and requested expanded narrative detail.				
2	Complete	Continued from the previous chat. Sequentially requested activities exploring sound physics, mathematics, and gravity vs. friction. Ultimately selected and formatted a trigonometry-based inclination activity without further extension.	No	14	5	0
3	Complete	Provided a comprehensive description of the desired physical activity parameters. Selected an idea, requested clear instructions, and added a specific student challenge.	No (Formatting only)	10	5	2
1	Complete	Ignored guiding questions. Selected an idea, explicitly requested narrative and physical components, and requested the solution draft.	No (Formatting only)	14	5	2
2	Complete	Uploaded a pre-existing AI-drafted document alongside the template.	No (Formatting only and	8	12	1

		Requested visual mockup descriptions, explored physical intervention methods to simulate scenarios on the mockup, and requested a final compiled document.	image generation not used)			
3	Complete	Requested an inclination technology activity. Selected an idea for development. Requested context, instructions, and an image representation of Protobject blocks (resulting in a text description).	No (Formatting only)	20	8	2
1	Empty	No chat log available.	N/A	N/A	N/A	N/A
2	Complete	Instead of selecting from generated ideas, the user dictated specific devices and tone, forcing the AI to develop a specific function classification activity aligned with those constraints.	No (Formatting only)	2	3	0
3	Complete	Requested ideas for a specific unit. Generated a second batch of 5 ideas, selected one, and requested instructions formatted strictly as a structured class plan, followed by a summary.	No (Formatting only)	14	7	3

1	Complete (Invalid)	Utilized unavailable legacy features (Deep Research / Gemini Whiteboard). Data invalid.	N/A	N/A	N/A	N/A
2	Incorrect	Iteratively requested motivational narratives for seismic movements, ignoring the AI's structural proposals. Ultimately selected an idea but filled the final template incorrectly.	N/A	N/A	N/A	N/A
3	Complete	Provided specific unit contents. Suggested and then rejected an "energy" theme. Selected a non-energy idea and requested clearer instructional steps.	No (Formatting only)	17	5	1

3. Human-Human Condition

Round	Activity Status	Interaction Summary	Time (min)
1	Complete	The participants debated whether to frame the activity around Mathematics or Chemistry, noting their preference for direct data since they do not consider themselves "humanist". They	20

		decided on a transversal approach focused on gathering noise level data to graph, which they checked if it was possible on protobject, and interpret. They experienced some hesitation formulating guiding questions , but agreed to use a 10-minute noise measurement visualized by a color-coded "traffic light" lamp system. During the design process, they were briefly distracted by off-topic joking.	
2	Complete	The participants collaborated to create a Technology and Physics activity analyzing series and parallel circuits, aligning it with the International Baccalaureate (IB) program. The group lost significant time on a tangent discussing the IB program's implementation difficulties and heavy parent involvement. Because they lacked specific physical components like resistors, they creatively used the inclination sensor to represent electrical resistance, framing it within a narrative of helping a neighbor at a social camp whose lights are malfunctioning. They also engaged in off-topic chatter about other teachers and schools	26
3	Complete	Utilized Gemini AI outside the designated framework (excluded from core analysis).	10
1	Complete	The participants designed a middle school activity focusing on classifying natural resources across Chile's macro-zones. They recalled past Kahoot-style activities , deciding to use the inclination sensor so students could tilt their phones left for "non-renewable" and right for "renewable" resources. The team completely went off-topic to discuss personal travel, architecture, and the church of Chillán. Despite	17

		this, they successfully established game mechanics and tried to play it in protobject like restarting upon a mistake and passing the phone, with extensions including a hit counter and a prize for completing all zones.	
2	Almost Complete (Missing Title)	The group aimed to teach computational thinking in mathematics. However, the participants got heavily side-tracked discussing the IB program, international travel opportunities for students, and the associated costs. They spent considerable time discussing personal commutes, the metro, and moving closer to the school. Additionally, they lost time discussing how to create and train custom AI "gems" instead of focusing on the manual activity design, leaving the final written artifact somewhat disjointed.	26
3	Complete	The participants designed an interdisciplinary Math and Physics activity for high school students focusing on calculating speed using a phone's accelerometer, which they physically tried using in protobject. They debated whether the focus should lean more towards math or physics. They spent time planning the real-world logistics of how students would ask permission to leave the classroom to measure the speed of younger students. Students would then compare manually calculated speeds to the accelerometer data. The session concluded with a philosophical tangent about Zeno's paradox of the turtle.	23
1	Complete	The group focused on designing a noise pollution activity for first-year high school students. They started with confusion regarding the template and what specific objective they	17

		were supposed to achieve. They lost time debating the exact decibel ranges, struggling to remember the specific thresholds for normal noise, danger, and pain (120 decibels). They also spent time discussing the practical logistics of testing the noise levels during school recess. Ultimately, they decided to use an audio reproducer to generate alerts and a 7x7 drawing to display different colors based on the sound range.	
2	Complete	The group came up with the idea of a navigation activity using a compass, trying the protobject compass, creating a narrative about a pirate ship crew hiding a treasure. Participants expressed difficulty and lost time adjusting to creating the activity entirely "by hand" without AI assistance. They went on a tangent discussing the difference between magnetic north and true north, and how maps are physically oriented. They also had an off-topic conversation about sharing the cost of a ChatGPT Plus account with a colleague. The activity's goal was for one group to hide an object and write specific step-by-step instructions for another group to follow.	26
3	Almost Complete (Missing Extension)	The participants aimed to create a mathematics and physics activity using the phone's accelerometer to calculate running speed based on time and distance. They tested the functionality in protobject. They were briefly distracted trying to figure out if they were allowed to use AI generators for this round. They spent time comparing manual creation to AI, noting that the AI formats things better and makes the code easier to read. Students would gather data to build a comparative table, formulate hypotheses, and analyze statistics ,	20

		and the session ended with off-topic banter about attending universities like Católica versus Chile.	

4. Overall Activity Summary

- **RAG Condition:** 16 Total Activities
- **Standard Condition:** 16 Activities (excluding multiple activities within a single round), or 18 Activities (inclusive of multi-activity generation).
- **Human-to-Human Condition:** 6 Activities 100% complete; 8 total activities in evaluable condition.